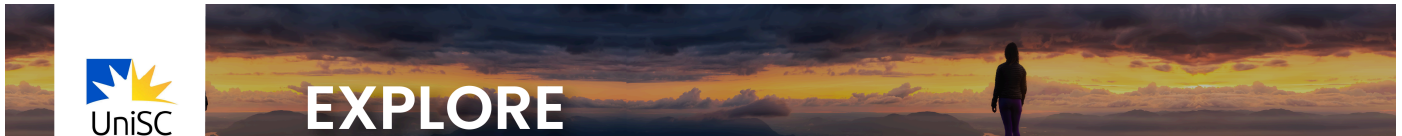


6.1 Projecting Costs and Revenues



Guiding Questions

1. What costs stay the same even if output changes?
2. Which costs rise when more units are sold?
3. How do revenue and cost assumptions affect profit?
4. At what sales level does the project break even?



What you will learn

- How to distinguish fixed and variable costs in a project
- How to estimate revenue using price and quantity assumptions
- How to calculate contribution margin and break-even point
- How to construct a simple profit projection model

A financial feasibility study starts with realistic forecasts of how much a project will earn and how much it will cost. The main purpose is to estimate whether the project can cover its costs, generate profit, and remain viable under reasonable assumptions.

Financial projections provide a structured way to assess whether a project is financially viable. The key idea is to translate assumptions about market conditions into expected revenues and costs.

A basic profit identity underpins the analysis:

Profit = Revenue - Variable Costs - Fixed Costs

The quality of the projection depends entirely on the realism and consistency of the underlying assumptions.

Cost structure: fixed vs variable

Fixed costs: Fixed costs do not change with output in the short run. They must be paid regardless of sales volume.

Examples:

- Rent
- Permanent staff salaries
- Insurance
- Software subscriptions
- Depreciation

Interpretation: These costs define the financial “baseline burden” of the project.

Variable costs: Variable costs change directly with output or sales volume.

Examples:

- Raw materials
- Packaging
- Payment processing fees
- Delivery or shipping
- Sales commissions

Interpretation: These determine the marginal cost of producing or selling one additional unit.

Revenue estimation**Revenue = Price per Unit x Quantity Sold**

In practice, the key challenge is not the formula itself but the assumptions:

How many units can realistically be sold?

Is demand stable, growing, or seasonal?

Will price remain constant under competition?

Contribution margin: The contribution margin measures how much each unit contributes toward covering fixed costs and generating profit:

Contribution Margin = Selling Price - Variable Cost

A higher contribution margin means:

- Faster recovery of fixed costs

- Higher profitability once break-even is reached
- Greater sensitivity of profit to sales increases

Break-even analysis: The break-even point is the level of sales at which total profit equals zero:

Break-even Quantity = Fixed Cost/Contribution Margin

This is a critical indicator of project risk:

Below break-even → losses

Above break-even → profits

Example: Online Tutoring Business

Step 1: Assumptions

Selling price per package = \$200

Variable cost per package = \$60

Annual fixed costs = \$28,000

Step 2: Contribution margin

Each sale contributes \$140 toward fixed costs and profit.

Step 3: Break-even point

Interpretation: The business must sell 200 packages per year to avoid losses.

Step 4: Profit scenarios

Scenario A: Low sales (180 packages)

Revenue = $180 \times 200 = 36,000$

Variable costs = $180 \times 60 = 10,800$

→ Loss of \$2,800

Scenario B: Base case (260 packages)

Revenue = $260 \times 200 = 52,000$

Variable costs = $260 \times 60 = 15,600$

→ Profit of \$8,400

This simple projection highlights three key economic insights:

- Break-even sensitivity: Small deviations around break-even can shift outcomes from profit to loss.
- Operating leverage: Once fixed costs are covered, additional sales generate relatively high incremental profit.

- Importance of demand assumptions: Project viability depends primarily on whether realistic sales volumes exceed the break-even threshold.

Although simplified, this type of deterministic forecasting is widely used in early-stage feasibility analysis because it provides a transparent link between assumptions and financial outcomes. It is particularly useful for evaluating project viability before more advanced modelling or empirical forecasting techniques are introduced.